Critical Thinking Rubric
Weight of assessment in final grade: 20%
Type of assessment: individual
Evaluator: Director CSBB

	Learning Outcome (numbers based on overall outcomes)	Criteria	Fail	Pass	Comments
Collaborate	2.c. Identify problems and formulate solutions to collaboration-related problems.	Demonstrates ability to reflect on group work and addresses challenges in team communication or coordination.	Fails to identify key problems in collaboration or suggests vague, unrealistic, or no solutions. Shows limited understanding of collaborative processes. No evidence of reflection on group processes.	Identifies relevant collaboration problems and proposes practical, constructive solutions that show understanding of team dynamics. Shows basic reflection on team dynamics or communication	
	2.d. Understand ethics and standard practice around credit and ownership in scientific group work.	- The way the team worked together (collaboration, cooperation or coordination) - The role(s) of the student in the team and project	The critical thinking journal lacks reflection on any of the following points: - The way the team worked together (collaboration, cooperation or coordination) - The role(s) of the student in the team Or the reflection is superficial and only scratches the surface.	The critical thinking journal contains detailed reflections on all the following points: - The way the team worked together (collaboration, cooperation or coordination) - The role(s) of the student in the team - Giving credit	
Research	3.b. Critically evaluate research strategies and if necessary, modify and adapt these strategies	Research insight	Fails to critically think about research processes and methods.	is able to think critically about methods being used including suggesting alternatives.	
	3.d. Analyse outcome of an experiment in a critical way	Data analysis	Struggles with identifying problems, did not show ability to critically think about data analysis methodologies	Identifies problems, has long-term vision, and reflects on challenges in data analysis.	
	3.e apply research ethics around reproducibility, data management and integrity	Research ethics	Struggles with identifying problems. Did not show understanding of ethics in reproducibility, data management and integrity.	Understands problems, is able to demonstrate understanding of ethics in reproducibility, data management and integrity.	
	3.f Understand processes involved with institutional review, grant applications and scientific publications	External review	Does not demonstrate understanding of scientific review processes	Understands processes of scientific review including research integrity, grants and publications	
	3.g. Apply ethical and philosophical considerations in performing research	Able to put research in ethical and philosophical context	Does not demonstrate understanding of ethical and philosophical considerations in research.	Identifies problems, has long-term vision, and reflects on ethical and philosophical context of research.	
Reflect	4.a. Use provided frameworks to reflect on experiences in minor (including mistakes that were made)	Describe your experience and what mistakes you identified within CSBB minor and what could be learned from them.	No or hardly any reflection is provided on experiences or mistakes made in the minor.	Reflection on experiences and mistakes in the minor are provided and it has been indicated what can be learned from the mistakes.	
	4.b. Understand and apply the concept of perspective agility	Use and reflect on different perspectives. Describe the value of perspective agility.	No or hardly any reflection is provided on the benefit of perspective agility.	Reflection is provided on perspective agility and what benefit it offered.	
	4.c. Understand how personal biases influence work.	Describe what personal and cultural preferences influences collaboration and scientific research.	Student shows no or poor understanding of biases in relation to collaboration and science.	Student can formulate and is aware of biases how that (might) influence collaboration and science.	

FINAL GRADE	PASS or FAIL